Regressionsmodeller

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Hvorfor ønsker vi regressionsmodeller?

- Indtil nu:
  - Estimater og beregninger for forskel mellem (mest to) grupper
- Ofte har vi brug for noget mere kompliceret
  - Mere end to grupper
  - Numeriske eksponeringer
  - Komplicerede udfald (f.eks. rater)
- Desuden ønske om at justere
  - for at tage højde om konfundering (mere på torsdag)
  - for at prædiktere mere præcise (sidst i kurset)
  - for at opnå mere styrke i tests / smallere konfidensintervaller
- Løsningen er ofte regressionsmodeller

(Simpel) Lineær regression

- Den simpleste regressionmodel vi bruger
- Numerisk udfald Y og eksponering X

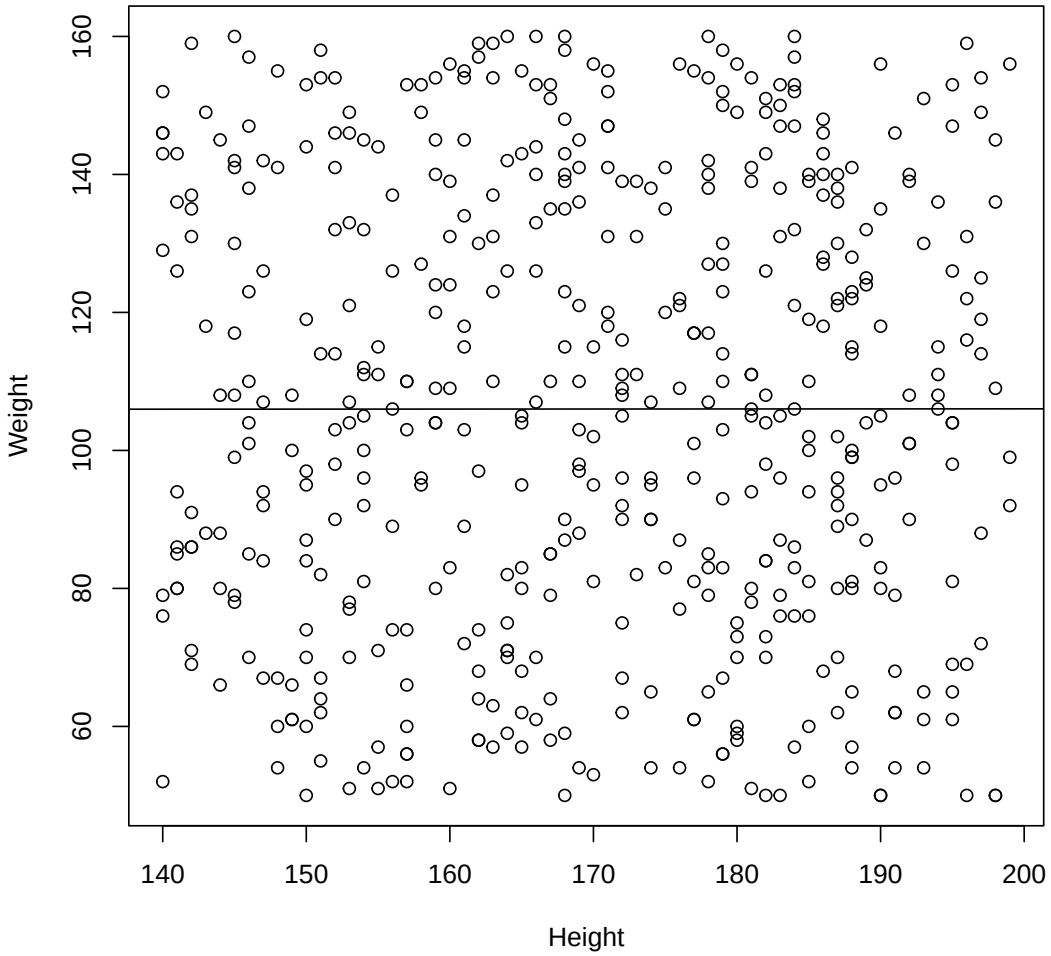
$$Y = \beta_1 X + \beta_0 + \epsilon$$

- $\epsilon \sim N(0, \sigma^2)$ normalfordelt med konstant spredning
- β_0 gennemsnit værdi for Y for personer med $X = 0$
- β_1 gennemsnitlig stigning (eller fald) i Y for 1 enheds stigning i X
- Nogle modelantagelser (nærmere om disse i næste uge)

Simple lineær regression i R

```
> library(readr)
> bmi <- read_csv("bmi.csv")
> linreg <- lm(Weight ~ Height, data=bmi)
> linreg
Call:
lm(formula = Weight ~ Height, data = bmi)
Coefficients:
(Intercept)       Height
  1.059e+02    8.819e-04
> coef(linreg)
 (Intercept)       Height
1.058501e+02 8.818709e-04
> summary(linreg)
Call:
lm(formula = Weight ~ Height, data = bmi)
Residuals:
    Min      1Q  Median      3Q     Max
-56.025 -26.016  -0.017  29.989  54.022
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 1.059e+02  1.513e+01   6.996 8.51e-12 ***
Height      8.819e-04  8.862e-02   0.010    0.992
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
Residual standard error: 32.42 on 498 degrees of freedom
Multiple R-squared:  1.989e-07,        Adjusted R-squared:  -0.002008
F-statistic: 9.904e-05 on 1 and 498 DF,  p-value: 0.9921
> confint(linreg)
                 2.5 %      97.5 %
(Intercept) 76.1250938 135.5751689
Height      -0.1732241   0.1749879
```

Figur: Simple lineær regression

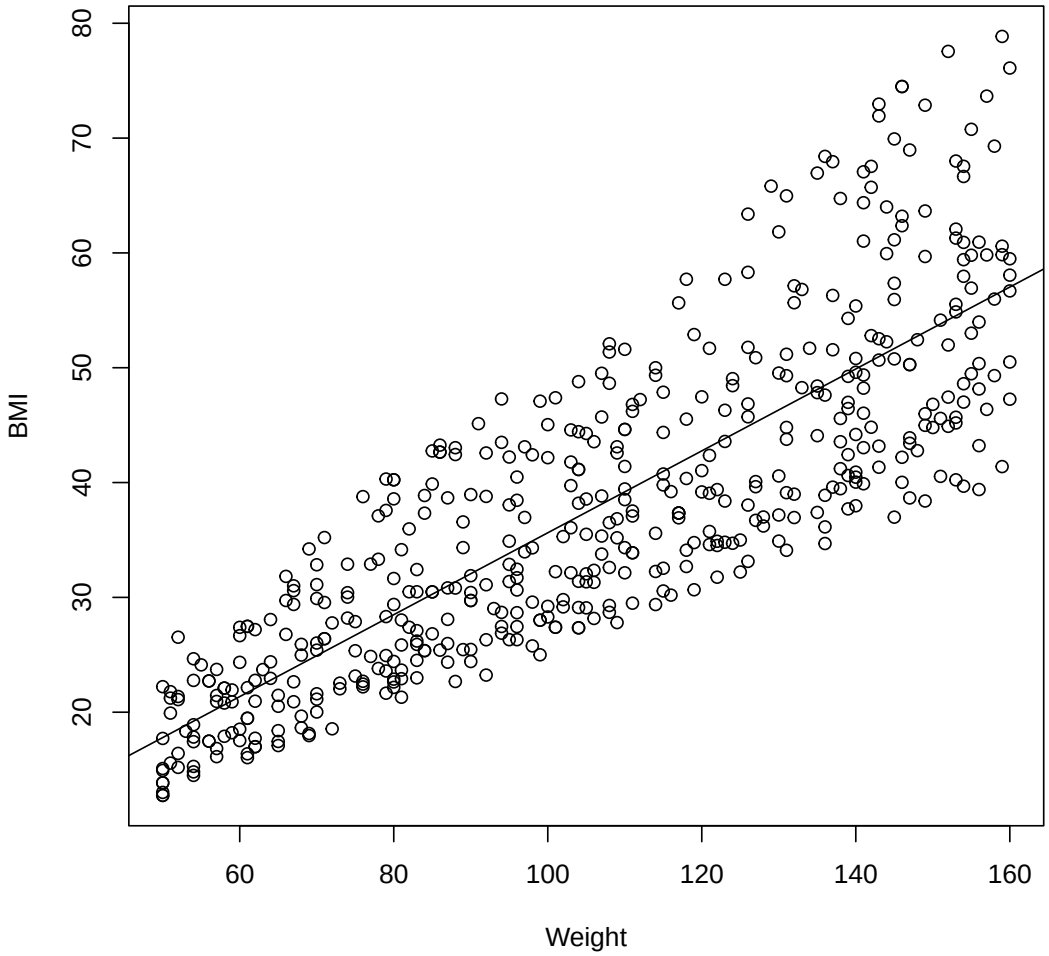



```
> pdf("linreg_Weight_Height.pdf")
> plot(Weight ~ Height, data=bmi)
> abline(linreg)
> dev.off()
```

Simple lineær regression i R 2

```
> linreg <- lm(BMI ~ Weight, data=bmi)
> linreg
Call:
lm(formula = BMI ~ Weight, data = bmi)
Coefficients:
(Intercept)       Weight
   -0.04163      0.35667
> coef(linreg)
(Intercept)      Weight
-0.04163307  0.35666903
> summary(linreg)
Call:
lm(formula = BMI ~ Weight, data = bmi)
Residuals:
    Min      1Q  Median      3Q     Max
-16.206  -5.717  -1.195   5.161  23.379
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.04163    1.20405  -0.035    0.972
Weight       0.35667    0.01086  32.829   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
Residual standard error: 7.859 on 498 degrees of freedom
Multiple R-squared:  0.684,        Adjusted R-squared:  0.6833
F-statistic:  1078 on 1 and 498 DF,  p-value: < 2.2e-16
> confint(linreg)
                 2.5 %    97.5 %
(Intercept) -2.4072810 2.3240149
Weight       0.3353235 0.3780145
```

Figur: Simple lineær regression 2




```
> pdf("linreg_BMI_Weight.pdf")
> plot(BMI ~ Weight, data=bmi)
> abline(linreg)
> dev.off()
```

Multipel lineær regression

- Udvidelse med flere forklarende variable
- Numerisk udfald Y og forklarende variable $X_1, \ldots, X_k$

$$Y = \beta_1 X_1 + \ldots \beta_k X_k + \beta_0 + \epsilon$$

- $\epsilon \sim N(0, \sigma^2)$ normalfordelt med konstant spredning
- β_0 gennemsnit værdi for Y for personer med $X = 0$
- β_k gennemsnitlig stigning (eller fald) i Y for 1 enheds stigning i X_k
- Når alle andre X_j holdes konstant

Multipel lineær regression i R

```
> linreg <- lm(Weight ~ Height+factor(Gender), data=bmi)
> linreg
Call:
lm(formula = Weight ~ Height + factor(Gender), data = bmi)
Coefficients:
       (Intercept)              Height  factor(Gender)Male
         1.055e+02           1.215e-03           6.169e-01
> coef(linreg)
       (Intercept)             Height factor(Gender)Male
      1.054912e+02       1.215137e-03       6.169494e-01
> summary(linreg)
Call:
lm(formula = Weight ~ Height + factor(Gender), data = bmi)
Residuals:
    Min      1Q  Median      3Q     Max
-56.349 -26.336  -0.029  29.833  54.310
Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)        1.055e+02  1.524e+01   6.923 1.37e-11 ***
Height             1.215e-03  8.871e-02   0.014    0.989
factor(Gender)Male 6.169e-01  2.903e+00   0.213    0.832
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
Residual standard error: 32.45 on 497 degrees of freedom
Multiple R-squared:  9.106e-05,        Adjusted R-squared:  -0.003933
F-statistic: 0.02263 on 2 and 497 DF,  p-value: 0.9776
> confint(linreg)
                        2.5 %      97.5 %
(Intercept)        75.5529777 135.4294015
Height             -0.1730861   0.1755164
factor(Gender)Male -5.0869367   6.3208355
```

Generaliseret lineær regression

- Vi ønsker også at håndtere ikke-kontinuerte udfald
- Derfor udvider vi lineær regression til generaliseret lineær regression (GLM)

$$g(Y) = \beta_1 X_1 + \ldots \beta_k X_k + \beta_0 + \tilde{\epsilon}$$

- g er en link-funktion, afhængig af modellen
- $\tilde{\epsilon}$ har ikke nødvendigvis en pæn fordeling
- I stedet for specificerer vi også en sandsynlighedsfordelingsfamile (eksempler om lidt)
- β_1 gennemsnitlig stigning (eller fald) i $g(Y)$ for 1 enheds stigning i X
- Den præcise fortolkning afhlænger af link-funktion og fordelingsfamile (eksempler om lidt)

Logistisk regression

- Dikotomt udfald, 0 eller 1

$$g(Y) = \log\left(\frac{Y}{1-Y}\right)$$

- g er logit-funktionen
- Fordelingsfamilie: Binomial / Bernoulli
- β_j er logodds for 1 enheds stigning i X_j
- $OR_j = exp(\beta_j)$ er odds ratio
- $OR_0 = exp(\beta_0)$ er odds når alle $X_j = 0$

Logistisk regression i R

```
> library(readr)
> diabetes <- read_csv("diabetes.csv")
> logreg <- glm(factor(Outcome) ~ Glucose, data = diabetes, family = "binomial")
> summary(logreg)
Call:
glm(formula = factor(Outcome) ~ Glucose, family = "binomial",
    data = diabetes)
Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -5.350080   0.420827  -12.71   <2e-16 ***
Glucose      0.037873   0.003252   11.65   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
(Dispersion parameter for binomial family taken to be 1)
    Null deviance: 993.48  on 767  degrees of freedom
Residual deviance: 808.72  on 766  degrees of freedom
AIC: 812.72
Number of Fisher Scoring iterations: 4
> exp(coef(logreg))
(Intercept)     Glucose
0.004747769 1.038599360
> exp(confint(logreg))
Waiting for profiling to be done...
                  2.5 %    97.5 %
(Intercept) 0.002030101 0.0105859
Glucose     1.032181592 1.0454386
```

Modeltjek logistisk regression i R

```
> library("performance")
> performance_hosmer(logreg, n_bins = 10)
# Hosmer-Lemeshow Goodness-of-Fit Test

  Chi-squared: 3.195
           df: 8
      p-value: 0.922

Summary: model seems to fit well.
>
> library(rms)
> logreg.res <- lrm(factor(Outcome) ~ Glucose, data =  diabetes, y = TRUE, x = TRUE)
> residuals(logreg.res, type = "gof")
Sum of squared errors     Expected value|H0                    SD                     Z
         132.57002132          134.31487580            0.60734464           -2.87292314
          P
        0.00406693
```

Hosmer-Lemenshow er den klassisk mest udbredte, men meget følsom over for antal kasser (bins). Derfor anbefales den nederste ("omnibustesten").

Log-binomial regression

- Dikotomt udfald, 0 eller 1

$$g(Y) = \log(Y)$$

- g er den naturlige logaritme
- Fordelingsfamilie: Binomial / Bernoulli
- β_j er log-relativ risiko for 1 enheds stigning i X_j
- $RR_j = exp(\beta_j)$ er relativ risiko
- $RR_0 = exp(\beta_0)$ er risiko når alle $X_j = 0$
- Modellen er numerisk udfordret, hvis nogle personer har meget lav/høj sandsynlighed
- Modellen kan give sandsynligheder større end 1 (uønsket)
- Disse udfordringer kræver brug af speciel R-pakke (logbin)

Log-binomial regression i R

```
> #install.packages("logbin")
> library(logbin)
> logbinreg <- logbin(factor(Outcome) ~ Glucose, data = diabetes, method = "em")
> summary(logbinreg)
Call:
logbin(formula = factor(Outcome) ~ Glucose, data = diabetes,
    method = "em")
Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.5840  -0.7953  -0.6543   1.0596   2.4729
Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -3.0575597  0.1658017  -18.44   <2e-16 ***
Glucose      0.0153372  0.0009448   16.23   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
    Null deviance: 993.48  on 767  degrees of freedom
Residual deviance: 827.58  on 766  degrees of freedom
  AIC: 831.58
AIC_c: 831.60
Number of iterations: 1040
> exp(coef(logbinreg))
(Intercept)     Glucose
 0.04700226  1.01545541
> exp(confint(logbinreg))
                 2.5 %     97.5 %
(Intercept) 0.03396159 0.06505032
Glucose     1.01357670 1.01733761
```

Poisson regression

- Udfaldet er et antal
- Mest brugt med incidensrater (eller prævalensrater)
- Hvor vi kender størrelsen på en eksponeret population

$$g(Y) = \log(Y)$$

- g er den naturlige logaritme
- Fordelingsfamilie: Poisson
- β_j er log-incidence rate ratio for 1 enheds stigning i X_j
- $IRR_j = exp(\beta_j)$ er incidence rate ratio
- $IRR_0 = exp(\beta_0)$ er incidence rate når alle $X_j = 0$
- Størrelsen på den eksponerede population skal inkluderes i modellen

Poisson regression i R

```
> library(readr)
> breastcancer <- read_csv("breastcancer.csv")
> breastcancer = subset(breastcancer,!is.na(Incidence))
> breastcancer$State = relevel(factor(breastcancer$Country),ref="California")
> modeltr <- glm(Incidence ~ State, family = poisson, offset = log(Population), data = breastcancer)
> exp(coef(modeltr))
       (Intercept)       StateAlabama        StateAlaska       StateArizona      StateArkansas
       0.001331477        1.100929028        0.900540179        0.999804321        1.037608506
...
> summary(modeltr)
Call:
glm(formula = Incidence ~ State, family = poisson, data = breastcancer,
    offset = log(Population))
Coefficients:
                     Estimate Std. Error   z value Pr(>|z|)
(Intercept)        -6.6214661  0.0027902 -2373.147  < 2e-16 ***
StateAlabama        0.0961544  0.0079151    12.148  < 2e-16 ***
StateAlaska        -0.1047605  0.0219994    -4.762 1.92e-06 ***
StateArizona       -0.0001957  0.0072665    -0.027 0.978514
StateArkansas       0.0369186  0.0101955     3.621 0.000293 ***
...
> exp(confint(modeltr))
Waiting for profiling to be done...
                         2.5 %       97.5 %
(Intercept)         0.001324209  0.001338772
StateAlabama        1.083948962  1.118107576
StateAlaska         0.862274867  0.939938737
StateArizona        0.985642394  1.014121303
StateArkansas       1.017022125  1.058491610
```

Spørgsmål?